

SYS 304

Trade Studies, Risk, and Decision Analysis

Applications to Systems Engineering

Systems Analysis Process

Systems Analysis

- ❑ Decision analysis is an important part of systems analysis.
- ❑ Studying systems analysis.
 - Mathematics can be studied for its intrinsic truth and beauty.
 - Systems analysis is different: it exists to solve real problems for real customers.
 - Tools of systems analysis are important, but they are not the product.
 - The product is obtained by using systems analysis tools for solving real-world problems.

Rules for Systems Analysis

- Rule 1: There always is a client.
 - It is easy to get subjective.
 - It is easy to carve the system problem to fit your tools and to suit yourself.
 - Client provides necessary input.
 - Client-focused analysis provides a reality check.

Rules for Systems Analysis

- ❑ Rule 2: Your client does not understand his own problem.
 - Client is part of the problem and knows that.
 - It is human nature for the client to attempt to cover up his mistakes.
 - Client lies accidentally or unconsciously even more often.
 - Systems analyst is called upon when there is a problem.
 - If the client knew what the problem is, he would fix it without seeking the analyst's advice.

Rules for Systems Analysis

- ❑ Rule 3: The original problem statement is too specific; you must generalize the problem to give it contextual integrity.
 - Discussions between analyst and client lead to a mutual understanding of the client's problem, which usually is too specific.
 - Analyst must generalize client's problem first to arrive at a better understanding of the client's needs.
 - Generalizing too far broadens the topic to the point that it is difficult to solve it in the time and with the money available.

Rules for Systems Analysis

- ❑ Rule 4: The client does not understand the concept of the index of performance.
 - System analysts go about optimizing things (maximize profit, maximize mean time between failures, minimize cost, minimize installation time).
 - Truth is, almost no one optimizes anything in real life.
 - In real life, we try to behave so as to avoid major risks.
 - In real life, we seek merely to better things incrementally.
 - The index of performance is the tool used for measuring the progress the system is making toward the optimum.

Rules for Systems Analysis

- ❑ **Rule 5: You are the analyst, not the decision-maker.**
 - Standard cycle of emotions through which system analysts put themselves as they deal with an assignment.
 - First is fear: the client organization is often well known and thought to be competent.
 - Second is arrogance: how could the client be so stupid as to get himself in this?
 - Third is the maturation: analyst begins to identify closely to client's problem. But remember, you are the analyst not the decision maker.

Rules for Systems Analysis

- ❑ **Rule 6: Meet the time deadline and the cost budget.**
 - Clients go into hysterics if they are told we need to generalize the problem and take a top-down approach, because they suspect this is a trick to avoid facing hard reality, and the study won't finish on time and as budgeted.
 - But, generalizing is way of coming to grips with the actual problem.
 - The problem probably is ill-defined, or the system analyst wouldn't have been hired in the first place.

Rules for Systems Analysis

- ❑ Rule 7: Take a goal-centered approach to the problem, not a technology-centered or chronological approach.
 - Chronological approach seems so logical and straightforward that it might be called part of the folk wisdom of mankind.
 - Chronological approach requires exploration of many blind alleys.
 - Technological approach tends to produce an artificially narrow options field and may result in exclusion of superior solutions.

Rules for Systems Analysis

- ❑ Rule 8: Non-users must be considered in the analysis and in the final recommendations.
 - System analyst should respect the value system of non-users and not project his/her value system on others.
 - Also, we have to recognize that non-users are sometimes unreasonable.

Rules for Systems Analysis

- ❑ Rule 9: Computer simulations are not a solve all solution.
 - You cannot get out of a computer model any more than you put into it.
 - This does not mean, however, that you can not learn some surprising things from a computer model.
 - All of the solutions of a mathematical relationship are implied by the relationship itself.
 - It does not mean that you know all of the solutions when you write the mathematical equation.
 - The same is true of a computer model.

Rules for Systems Analysis

- ❑ Rule 10: The role of decision-maker in public systems is often a confused one.
 - It is true that role titles are sometimes used interchangeably.
- ❑ Analyst
 - Professional who aids those involved with a problem to order the alternatives and decide on a course of action.
 - Should be neutral and objective and should have no stake in the outcome.

Rules for Systems Analysis

- ❑ Rule 10: The role of decision-maker in public systems is often a confused one.
- ❑ Client
 - Person or group with whom the analyst interacts during goal definition.
 - Receives final report and sometimes pays the bill.
 - Affected by the outcome and thus is a stakeholder.
 - May be empowered to choose from among the options and thus is a decision maker, but not always.
- ❑ Stakeholders
 - All those affected by the system.
 - May be users or non-users.
 - Need not be clients or decision makers.

Rules for Systems Analysis

- ❑ Rule 10: The role of decision-maker in public systems is often a confused one.
- ❑ Sponsor
 - Pays the bill.
 - May or may not be the client.
 - Usually a stakeholder.
- ❑ Decision maker
 - Chooses from among the options.
 - Must choose the particular index of performance.